DANIEL LEWIS

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EDUCATION

University of California, Merced **Aug 2021 – May 2025**
B.S. Mechanical Engineering, Cum Laude GPA: 3.7 / 4.0

EXPERIENCE

Lockheed Martin Space **Jul 2025 – Present**
Materials & Processes Engineer, Solar Array Manufacturing Center *Sunnyvale, CA*

- Root-caused production solar-cell debonds to residual solvent from a dielectric-film process change; program reverted it.
- Built a Python contact-area imaging rig that caught voids the lab missed and cut 3 weeks to 1 day; now certifies lots.
- Ran a 4-factor DOE on solar-cell repair bonding, isolated 3 significant factors, and wrote the spec now in production.
- Certified a new automated cell-bonding robot for the LM 400 line; ran the DOE that set its bond parameters.
- Machined lap-shear fixtures and tested 24 coupons of two film adhesives on CFRP; results set the flight bond process.
- Supporting laminate design and deploy/retract tests on LVSAT, a 10-person R&D team building a lunar solar array.
- Review GPS III, Next Gen GEO, LM 400, and smallsat drawings for material outgassing and thermal-cycling compliance.
- Qualified 50+ crimp contact, wire, and tool combinations for flight harness by pull test to NASA and IPC requirements.

Lockheed Martin Space **May 2024 – Jul 2025**
Electronic Materials & Processes Intern (summer, then part-time through senior year) *Littleton, CO*

- Reconciled 200,000+ part numbers across four Lockheed business units, enabling cross-division bulk purchasing.
- Wrote test plans comparing Vertrel and IPA flux cleaning on circuit cards for compatibility with coatings and adhesives.

Electrified Adventures **May 2023 – Aug 2023**
Engineering Intern *Orange, CA*

- Integrated battery packs, controllers, and three-phase motors into e-bike drivetrains; debugged faults with an oscilloscope.

INDEPENDENT PROJECTS

Thermally Actuated Morphing Radiator for the Lunar Night **Jul 2026 – Present**

- Designing a lotus-bud radiator that opens in lunar day to shed heat and seals shut at night, with no motors or power.
- Rejected three kirigami/origami prototypes; caught a wrong-grade (superelastic) nitinol order before buying two-way springs.
- Tested 14 Zn/Al/steel bimetal strips against Timoshenko's bimetal theory: 12 of 14 bent within 2x of prediction, most responsive at 0.1 to 0.2 mm steel. Peak deflection 10.7 mm at +120 °C vs. 31 mm nitinol stroke at 80 °C; chose nitinol.
- Building a half-scale single-spring bud and a full-size 8-petal, 8-spring prototype in printed PETG; all petals printed.

Two-Element Synthetic Aperture RF Imager **Feb 2025 – Sep 2026**

- Built a 2.4 GHz rig: two Vivaldi antennas on a 0.5 to 60 cm adjustable rail feeding a two-channel bladeRF 2.0 radio.
- Diagnosed unstable direction finding as phase ambiguity from wide spacing; switched to a sliding synthetic aperture.
- Reconstructed an angle map over a photo that located a Wi-Fi router placed at 0, 30, 45, and 90 degrees.
- Predecessor: HackRF One spectrum analyzer (2048-point FFT, 60+ segment FCC band classifier).

UNDERGRADUATE RESEARCH AND DESIGN

Non-Equilibrium Plasma & Fluids Lab, UC Merced **Apr 2023 – May 2025**
Research Assistant, PI: Dr. Venkattraman Ayyaswamy *Merced, CA*

- Ran a 50 kV plasma reactor fixing air nitrogen into water; treated water matched nitrogen-enriched soil in plant growth.

Aeroelastic Wing Optimization, UC Merced **Aug 2024 – May 2025**
Undergraduate Researcher, PI: Dr. Sachin Goyal *Merced, CA*

- Modeled the Pazy benchmark wing in HyperMesh/OptiStruct; stringers in place of solid material cut structural mass 44%.

Senior Capstone: Guayule Rubber Processing, UC Merced **Aug 2024 – May 2025**
Advisor: Dr. Kara McCloskey *Merced, CA*

- Processed raw guayule rubber into tensile and compression samples, adapting a deli slicer to cut repeatable 2 mm slices.

Molecular Motor Biophysics, UC Merced **Aug 2021 – May 2022**
Research Assistant, PI: Dr. Ajay Gopinathan *Merced, CA*

- Simulated kinesin-2 and OSM-3 cargo-transport blockage in Python, a mechanism studied in Alzheimer's research.

SKILLS & CERTIFICATION

Materials & Test	Adhesive qualification, DOE, lap-shear and tensile testing, microscopy, failure analysis, ASTM E595
Fabrication	Composite layup, fixture machining, FDM printing (PETG), flight harness crimping
RF Hardware	bladeRF 2.0, HackRF One, Vivaldi antenna arrays, phase interferometry, synthetic aperture imaging
Software	Python (NumPy, image processing, FFT/DSP), HyperMesh/OptiStruct
Cert. & Interests	IPC/WHMA-A-620; composites fabrication, deployable space structures, RF instruments, photography